Atividade 11.1. Explorando a tabela ARP no Kali Linux

```
Aplicativos  Google — Mozilla Firefox  root@kali: ~                    2026-03-25 19:22  aluno
root@kali: ~
Sessão  Ações  Editar  Exibir  Ajuda
(Run: "touch ~/.hushlogin" to hide this message)
┌──(root㉿kali)-[~]
└─# arp -a
ip-192-168-98-201.ec2.internal (192.168.98.201) em 12:00:e4:93:f4:97 [ether] em eth0
ip-192-168-98-2.ec2.internal (192.168.98.2) em 12:b0:4e:d7:76:21 [ether] em eth0
ip-192-168-98-1.ec2.internal (192.168.98.1) em 12:b0:4e:d7:76:21 [ether] em eth0

┌──(root㉿kali)-[~]
└─# ip -s -s neigh flush all
192.168.98.201 dev eth0 lladdr 12:00:e4:93:f4:97  ref 1 used 52/0/52probes 4 REACHABLE
192.168.98.2 dev eth0 lladdr 12:b0:4e:d7:76:21  ref 1 used 12/12/12probes 1 REACHABLE
192.168.98.1 dev eth0 lladdr 12:b0:4e:d7:76:21  used 29/29/13probes 1 STALE

*** Round 1, deleting 3 entries ***
*** Flush is complete after 1 round ***

┌──(root㉿kali)-[~]
└─# arp -a
ip-192-168-98-201.ec2.internal (192.168.98.201) em 12:00:e4:93:f4:97 [ether] em eth0
ip-192-168-98-2.ec2.internal (192.168.98.2) em 12:b0:4e:d7:76:21 [ether] em eth0

┌──(root㉿kali)-[~]
└─# arp -a
ip-192-168-98-201.ec2.internal (192.168.98.201) em 12:00:e4:93:f4:97 [ether] em eth0
ip-192-168-98-2.ec2.internal (192.168.98.2) em 12:b0:4e:d7:76:21 [ether] em eth0
ip-192-168-98-1.ec2.internal (192.168.98.1) em 12:b0:4e:d7:76:21 [ether] em eth0

┌──(root㉿kali)-[~]
└─#
```

Figura 1: Tabela ARP no Kali Linux após acesso externo, mostrando novamente a entrada do gateway da rede

Atividade 11.2. Implementando Alias de Interface de Rede no Kali Linux

```
Aplicativos  root@kali: ~                                   2026-03-25 19:26  aluno
root@kali: ~
Sessão  Ações  Editar  Exibir  Ajuda

┌──(root㉿kali)-[~]
└──(root㉿kali)-[~]
└─# ssh aluno@192.168.98.42
The authenticity of host '192.168.98.42 (192.168.98.42)' can't be established.
ED25519 key fingerprint is: SHA256:KQ/spvWObNxAVxbfp2S1QhxZUrE5RwsDpIjQnDFA2QQ
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.98.42' (ED25519) to the list of known hosts.
aluno@192.168.98.42's password:
Linux kali 6.16.8+kali-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.16.8-1kali1 (2025-09-24) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
┌─(Message from Kali developers)
│
│ This is a minimal installation of Kali Linux, you likely
│ want to install supplementary tools. Learn how:
│ ⇒ https://www.kali.org/docs/troubleshooting/common-minimum-setup/
│
│ This is a cloud installation of Kali Linux. Learn more about
│ the specificities of the various cloud images:
│ ⇒ https://www.kali.org/docs/troubleshooting/common-cloud-setup/
└─(Run: "touch ~/.hushlogin" to hide this message)
┌──(aluno㉿kali)-[~]
└─$ exit
sair
Connection to 192.168.98.42 closed.
```

Figura 2: Conexão SSH estabelecida utilizando o alias de interface configurado em 192.168.98.42

Atividade 11.3. Implementando MAC Cloning no Kali Linux

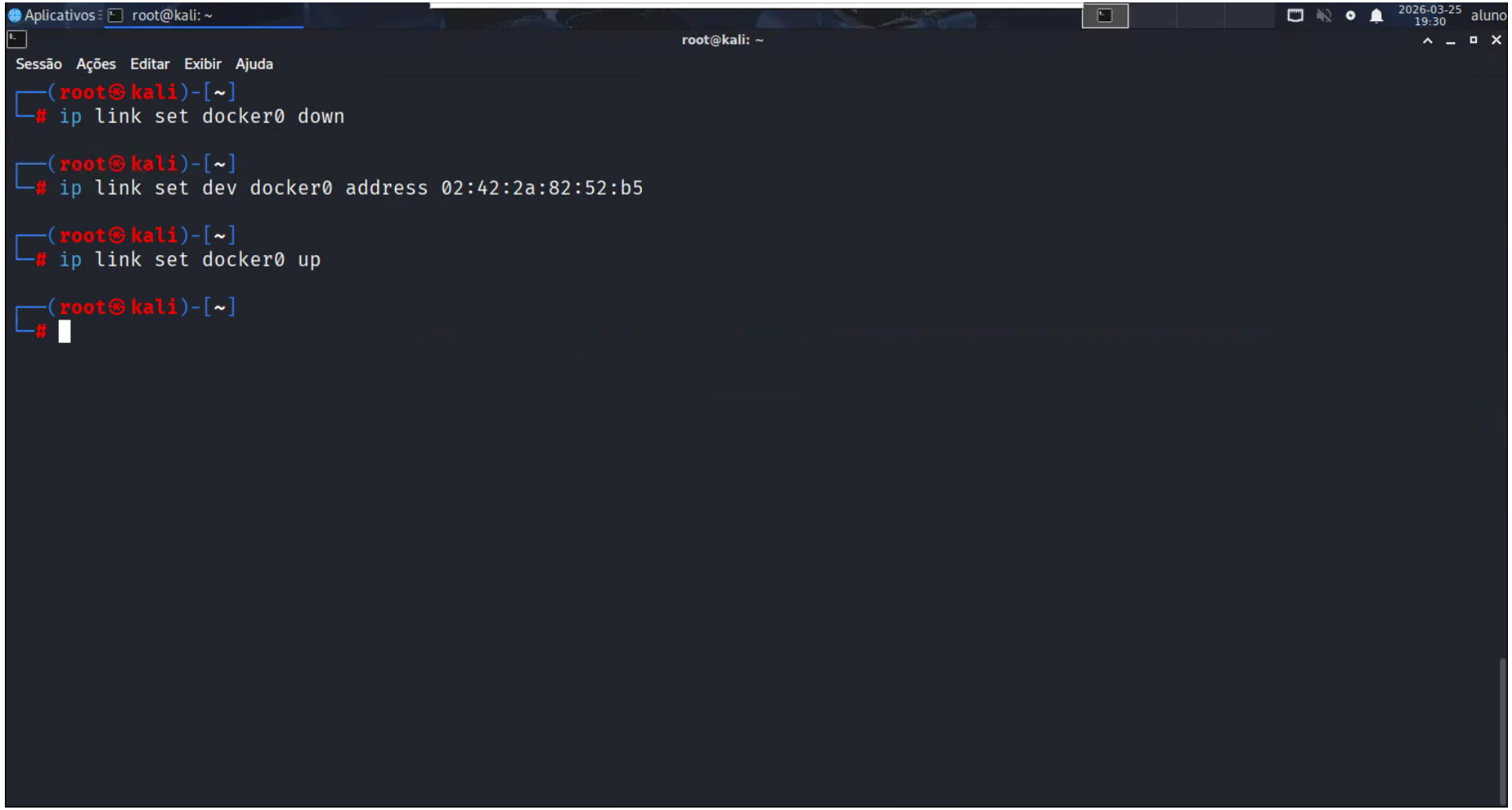

Figura 3: Reconfiguração da interface docker0 com retorno ao endereço MAC original

Atividade 11.4. Configurando QoS no Kali Linux

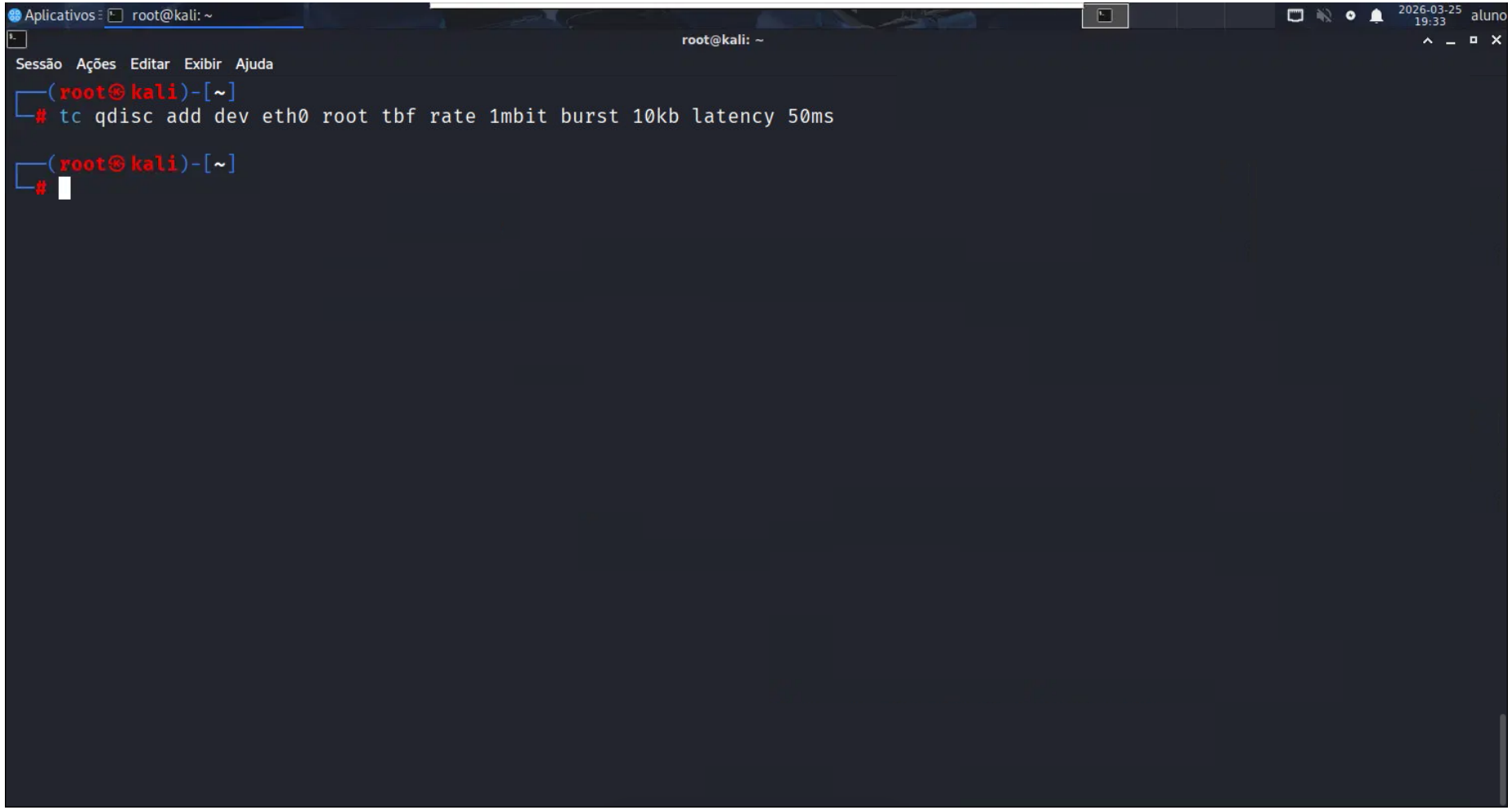

Figura 4: Aplicação de política de QoS com limitação de largura de banda via comando tc

Atividade 11.5. Configurando QoS no Windows Server 2022

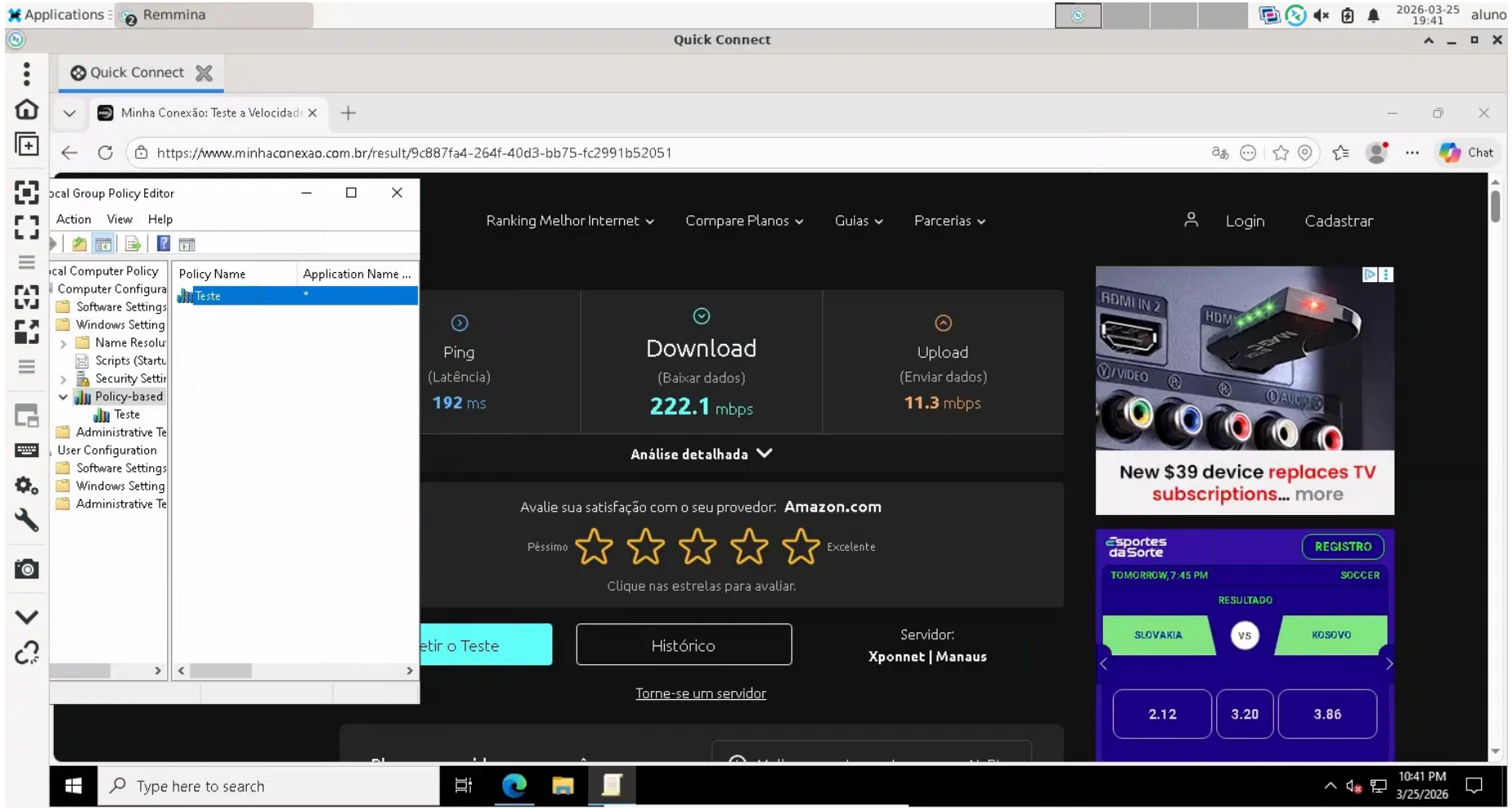

Figura 5: Teste de upload após aplicação da política de QoS com limitação de banda no Windows Server 2022